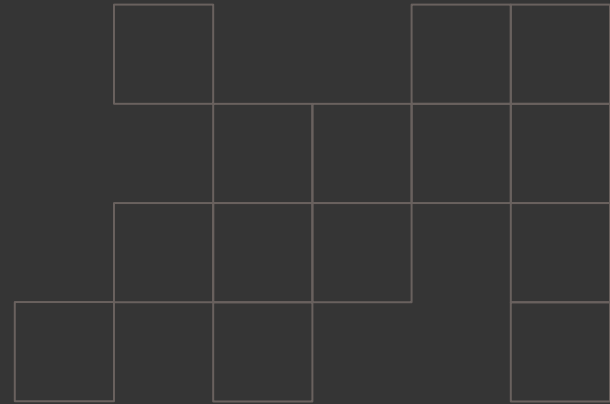
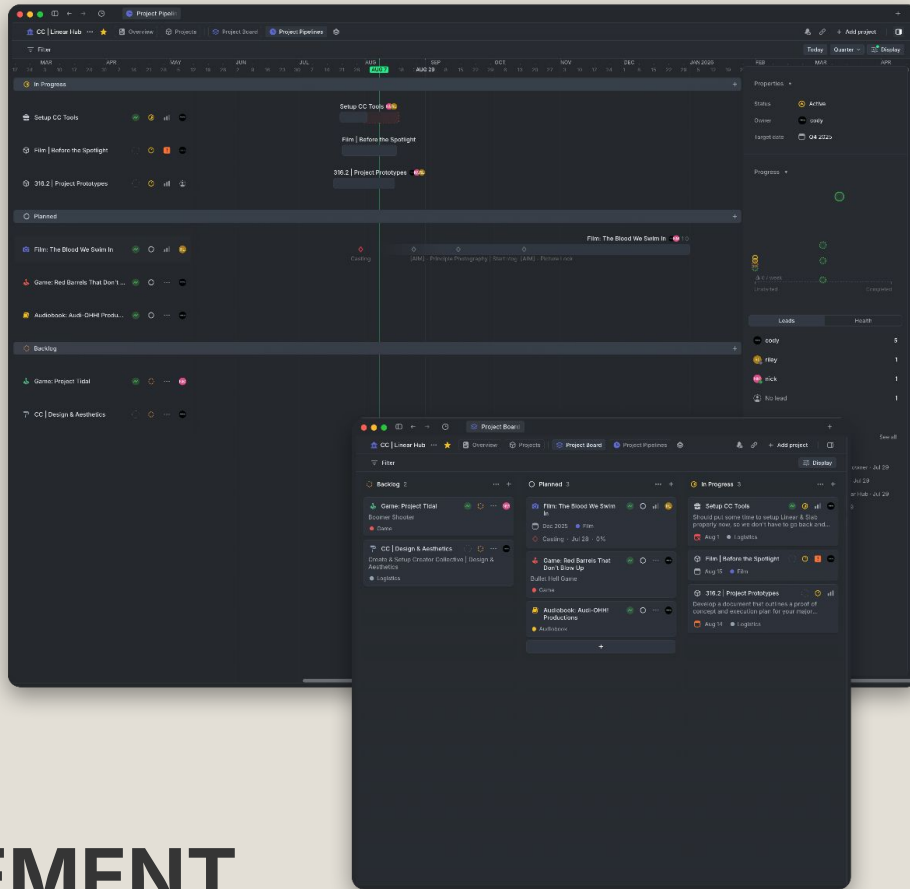
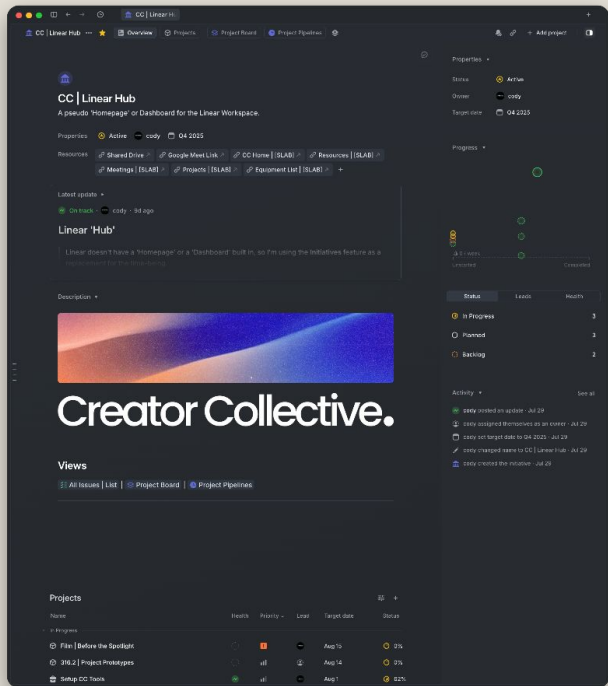


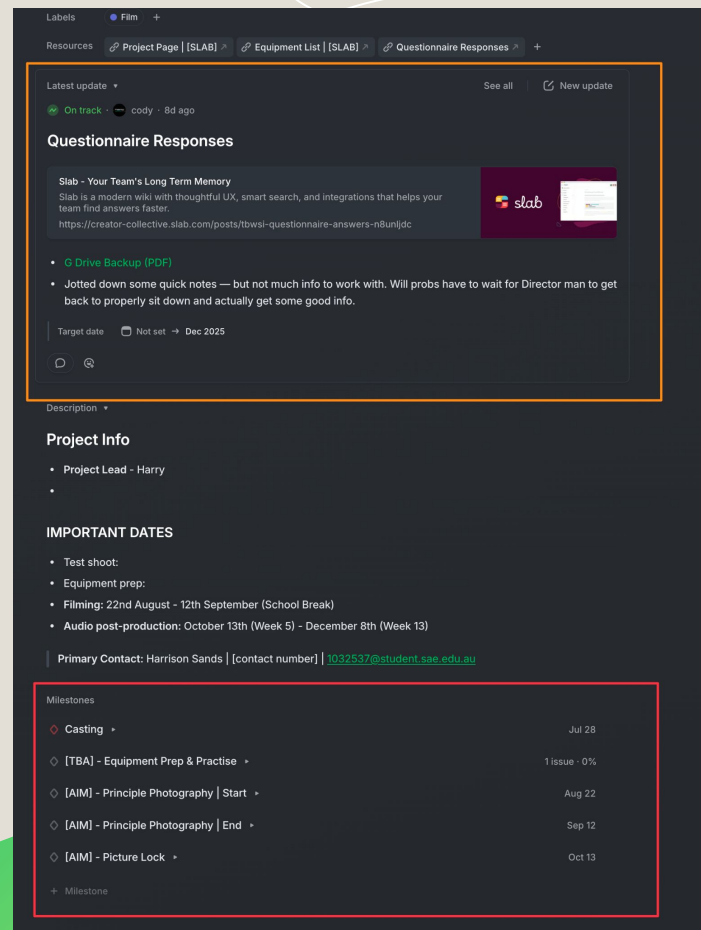
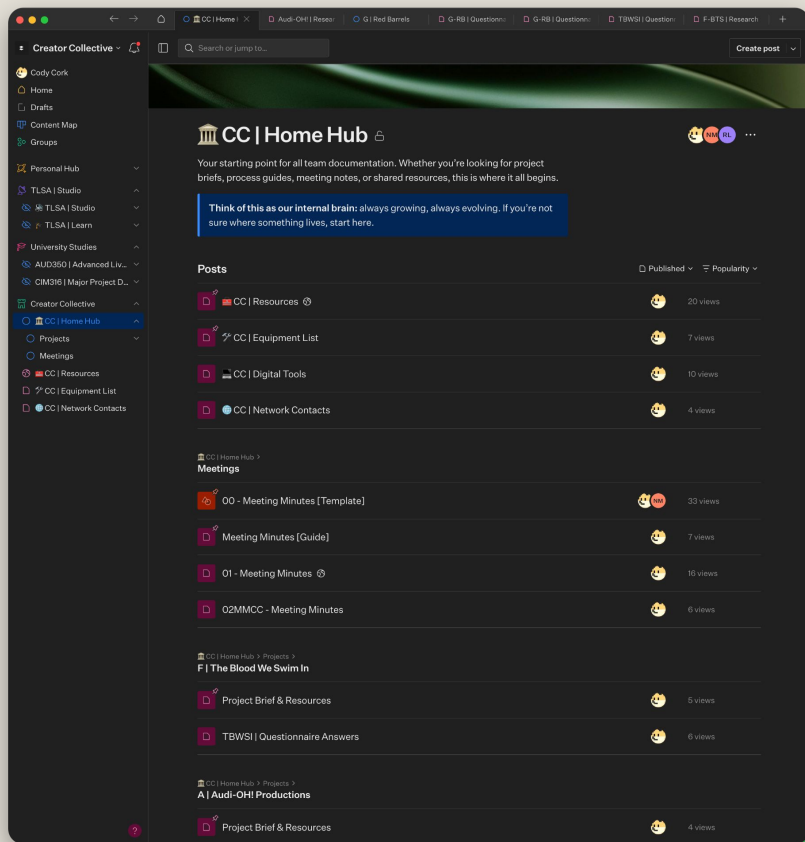
# Freelance Collective Prototype

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# PROJECT MANAGEMENT



Search

ChatGPT

GPTs

GptOracle | The AI Resea...

Life Coach Robin

DALL·E

New Project

316.2 | Research

Induction synonyms

New chat

Sound design research prompt

Research assistant instructions

Eco-brutalism synonyms

Free Miro FigJam alternatives

Music production course structure

ADR in film

Appointment date calculation

Track feedback summary

Summarize into bullet points

C#min vs F#min comparison

Summarize questionnaire points

Control F# F2 script

Resolution increase by 20%

Document tab apps

Key signature compatibility

Slab Wiki blurb

Bmin relative major

Cody Cork

Sound design in bullet hell

316.2 | Research

CREATIVE RESEARCH PROMPT: SOUND DESIGN IN BULLET HELL GAMES

1. CREATIVE CONTEXT & FRAMEWORK

- Expert(s) conducting the research: Senior Game Audio Designer, Psychoacoustic Researcher, Bullet Hell Game Developer, UI/UX Sound Interaction Specialist

- Ideation Domain: Interactive sound design for high-intensity games (bullet hell genre)

- Research Objective: Identify, analyze, and recommend best-in-class audio design practices that enhance player experience, responsiveness, and immersion in bullet hell games

- Current Understanding: Bullet hell games are characterized by overwhelming visual stimuli, rapid player decision-making, and precision control — sound design must support clarity, feedback, emotional regulation, and minimal auditory fatigue

- Knowledge Gaps & Opportunity: Lack of comprehensive guidelines for sound layering, psychoacoustic balance, auditory signaling, and fatigue prevention in chaotic visual environments

- Potential Impact: Improve game feel, reduce cognitive overload, enhance player flow state, and increase commercial/critical success through emotionally engaging audio design

2. CORE IDEATION QUESTION & HYPOTHESIS

- Primary Creative Question: What audio design strategies optimize clarity, emotional impact, and player feedback in bullet hell games without overwhelming the player?

- Working Hypothesis: Minimalist yet information-rich audio design — using tonal cues, spatial layering, and dynamic mix prioritization — maximizes player clarity and engagement in high-density visual environments

- Alternative Approaches: Maximalist retro soundscapes, procedurally adaptive audio layers, or AI-generated responsive audio design

- Theoretical Foundation: Grounded in flow theory, cognitive load theory, and auditory scene analysis (ASA)

3. METHODOLOGICAL PARAMETERS

- Creative Process: Design sprint analysis, gameplay sound breakdowns, prototyping in Unity/Unreal, player sound perception testing, reference mapping across iconic titles (e.g., Touhou, Cuphead, Ikaruga)

- Data Types: Sound spectrograms, user feedback logs, real-time audio telemetry, expert sound designer commentary

- Temporal Scope: Bullet hell titles released between 2000–2025 with focus on recent design innovations

- Sample Characteristics: Games rated highly for audio direction, indie and AAA spectrum, varying intensity levels and aesthetic themes

- Analytical Methods: Spectral layer decomposition, spatialization mapping, affordance analysis, emotional arc modeling

- Ethical/Compliance Protocols: None required, but inclusion of inclusive audio design principles encouraged

4. CREATIVE OUTPUT SPECIFICATIONS

- Report Structure: Design research dossier with diagrams, use-case breakdowns, annotated audio clips, and implementation recommendations

- Abstract Required: Yes

- Analytical Depth

- [ ] Level 1: Idea shortlist

Bullet Hell Sound Design Resea...

Annotated Bibliography: Sound Design in Bullet Hell & High-Intensity Games

1. Moldenhauer, M., & Veca, K. (2018, March). *Creating Cuphead Audio: Retro Cartoon Chaos*. Game Developers Conference.

Link: <https://www.gdcvault.com/play/1026090/Creating-Cuphead-Audio-Retro-Cartoon>

Source Tier: Tier 1 (Developer GDC Talk)

Summary: Explores the unique challenges of designing sound for Cuphead's chaotic gameplay and vintage aesthetic. Key insights include managing clarity through syncopated rhythms, balancing musical authenticity with functional feedback, and creative use of Foley to emphasize cartoon violence.

Relevance: Demonstrates how a heavily stylized, high-intensity game can maintain player clarity and emotional impact with a thoughtful audio mix. Highlights retro maximalism done right.

2. Berta, R., Bellotti, F., & De Gloria, A. (2021). *Psychoacoustics and Game Audio: A Review and Implications for Game Design*. *Frontiers in Psychology*, 12, 641227. <https://doi.org/10.3389/fpsyg.2021.641227>

Summary: This review paper examines how psychoacoustic principles (e.g., loudness perception, masking effects, frequency salience) apply to game audio design. It emphasizes how sound can support decision-making, emotional immersion, and sensory management in high-load situations like combat games.

Relevance: Provides a foundational framework for applying auditory masking theory and perceptual prioritization to bullet hell environments. Strong backing for dynamic mixing strategies and minimalist audio cues that reduce cognitive overload.

3. Houghton, A. (n.d.). *Going Beyond the Beep: A Look into Inaudible Sonic Feedback for Gameplay Enhancement*. Audiokinetic.

Link: [https://www.audiokinetic.com/library/2022/15\\_8156/?source=SDK&id=going\\_beyond\\_the\\_beep\\_a\\_look\\_into\\_inaudible\\_sonic\\_feedback\\_for\\_gameplay\\_enhancement.html](https://www.audiokinetic.com/library/2022/15_8156/?source=SDK&id=going_beyond_the_beep_a_look_into_inaudible_sonic_feedback_for_gameplay_enhancement.html)

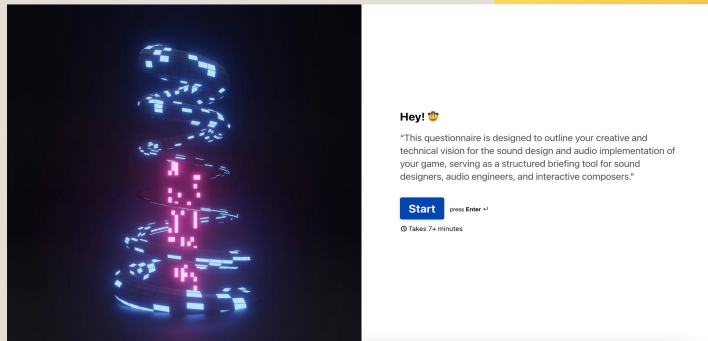
Summary: This piece explores how Wwise can be used to implement subliminal, non-intrusive sonic feedback that enhances player experience without overwhelming the soundscape. It discusses masking, frequency shaping, and dynamically triggered sub-threshold cues to aid gameplay perception.

Relevance: Highly applicable to bullet hell games where audio clarity must

# Project Onboarding Questionnaire

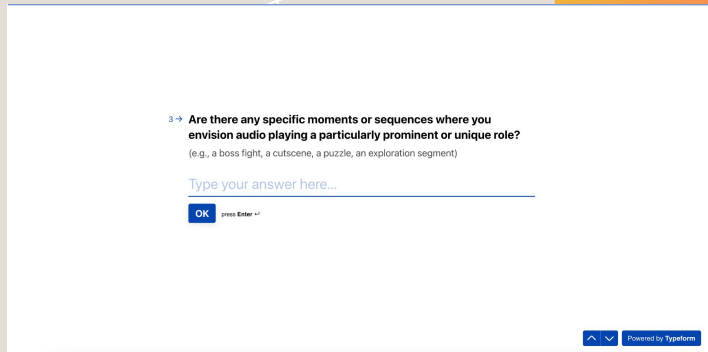
## INITIAL GOAL

- Questionnaire is used to help streamline the production process and outline project expectations and vision from an audio standpoint.
- Streamlined feedback process.



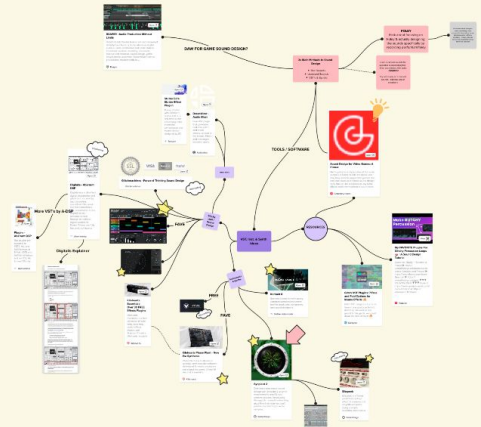
## WHAT WE'VE LEARNED

- Some questions were missed by us.
- Forgot to emphasise that the project leads need to be answering the questions.



BIG BRAINSTORM

## BRAINSTORM



### Ideas & Brain Dumps

### Research Brainstorm

Research Brainstorm

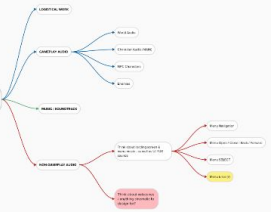
## RED BARRELS GAME



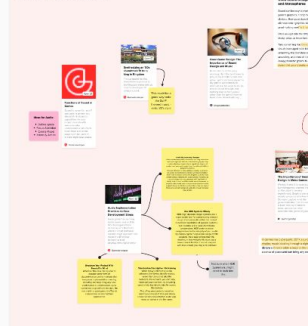
### Mindmap Colour Legend

|            |             |              |               |
|------------|-------------|--------------|---------------|
| <b>Red</b> | <b>Blue</b> | <b>Green</b> | <b>Yellow</b> |
| Gameplay   | Visuals     | Sound        | Story         |
| Marketing  | Development | Gameplay     | Visuals       |

### Idea Mind Map



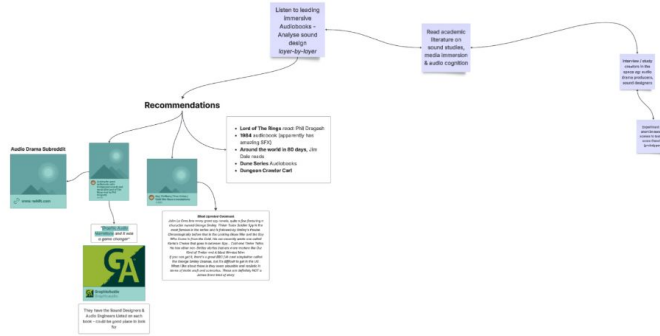
### Game Audio, Dev & Logistics



## Suggested Methodologies

### Suggested Methodologies

- Listen to leading immersive audiofiles and analyse the sound design team by team
- Read academic literature on sound studies, media immersion, and auditory cognition
- Experiment on sound perception in the sound field, audio drama production, sound design/production
- Experiment with short interactive stories to test your theories (from sound prototypes)



Design all of the content items & thoughts relevant to the project & how, which can be organized and sorted later



## Technical Research

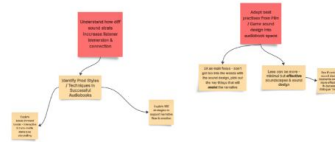


## Market / Audience Research

Headphones & Average Listening Behaviours / Expectations



## Production Styles & Techniques Research



BTS\_RoughCut\_1.mp4

01:04:08:25

04:19

11:10





# Prototype Research 2

## Early 2000s Game Sound Design

- Researching equipment & samples used.
- Techniques & approaches
- Limitations

Odyssey Audio Spreadsheet

**We request that you give credit when posting these findings.**

| Sound Source                                                     | Bank/Program       | Name             | OST Name (Dominant appearance only, check column F for) | File Names (main/game, then soundtrack, then smp) | Credit        | Notes    |
|------------------------------------------------------------------|--------------------|------------------|---------------------------------------------------------|---------------------------------------------------|---------------|----------|
| Confirmed matches/minor verification needed (patch numbers, etc) |                    |                  |                                                         |                                                   |               |          |
| Halo: Combat Evolved                                             |                    |                  |                                                         |                                                   |               |          |
| Kurzweil K2500                                                   | 74                 | Ripper Bass      | A Walk in the Woods                                     | mor1_c40_extraction_music                         | Yewlyn        |          |
| Roland JD-990                                                    | B: 57              | Dreaming Time    | A Walk in the Woods                                     | mor1_c40_extraction_music                         | Noelle / Booe | Main sy  |
| Roland XV-5080                                                   | PR-D 104           | Darkshine        | A Walk in the Woods                                     | mor1_c40_extraction_music                         | Booe          | Drony    |
| Roland XV-5080                                                   | PR-C 123           | Biosphere        | Alien Corridors                                         | scif1_full_scif1_lite                             | Neo           | Somew    |
| Roland XV-5080                                                   | PR-C 094           | Albion           | Ambient Wonder                                          | magic_a50_lft_music                               | Neo           |          |
| Roland XV-5080                                                   | PR-C 112           | Cascade          | Ambient Wonder                                          | magic_a50_lft_music                               | Neo           | Patch is |
| E-MU E3 More Emulator Standards                                  | Arco& Spic Celli&M | Velo Spiccatto   | Brothers in Arms                                        | battle1_battle1_themes_x10_music03_x30_music      | Booe          |          |
| E-MU E3 Emulator Standards                                       | Full Arco String   | Arco Strings Lo  | Brothers in Arms                                        | battle1_battle1_themes_x10_music03_x30_music      | Booe          |          |
| E-MU E3 Orchestral                                               | SoloEnglishHorn    | St English Horn  | Brothers in Arms                                        | battle1_battle1_themes_x10_music03_x30_music      | Booe          |          |
| E-MU E4 Ultra Production Set                                     | Orchestral         | Flute w/Vib      | Brothers in Arms                                        | battle1_battle1_themes_x10_music03_x30_music      | Booe          |          |
| E-MU E3 Orchestral                                               | Solo Fr Horns 8M   | ff Velo Horn     | Brothers in Arms                                        | battle1_battle1_themes_x10_music03_x30_music      | Booe          |          |
| E-MU E3 Orchestral                                               | Marcato Strgs 8M   | Velo Basses      | Brothers in Arms                                        | battle1_battle1_themes_x10_music03_x30_music      | Booe          |          |
| E-MU E3 Orchestral                                               | Boys Choir 8M      | Slow Attacks     | Brothers in Arms                                        | battle1_battle1_themes_x10_music03_x30_music      | Neo / Booe    |          |
| E-MU E3 Orchestral                                               | Solo Trombones 2   | ff Velobone      | Brothers in Arms                                        | battle1_battle1_themes_x10_music03_x30_music      | Booe          |          |
| E-MU E3 Orchestral                                               | Solo Trumpets 2 8M | St Trumpet Duet  | Brothers in Arms                                        | battle1_battle1_themes_x10_music03_x30_music      | Booe          |          |
| Roland XV-5080                                                   | PR-E 075           | Marcato Str      | Brothers in Arms                                        | battle1_battle1_themes_x10_music03_x30_music      | Booe          |          |
| E-MU Proteus 2000 (Protozoa)                                     | 082-0              | GulaGula         | Covenant Dance                                          | jig_jig_lite                                      | Booe          | FINALL   |
| Roland XV-5080                                                   | PR-D 102           | Soundtracue      | Covenant Dance                                          | jig                                               | Booe          | The sof  |
| Advanced Orchestra                                               | violins            | VI crescendo     | Devils...Monsters...                                    | nightmare                                         | Booe          |          |
| Advanced Orchestra                                               | violins            | VI sh. detache 1 | Devils...Monsters...                                    | nightmare                                         | Booe          |          |
| Advanced Orchestra                                               | violins            | VI sh. detache 2 | Devils...Monsters...                                    | nightmare                                         | Booe          |          |
| Roland XV-5080                                                   | PR-C 128           | Terminate        | Devils...Monsters...                                    | nightmare_nightmare_perc                          | Python Blue   | Very sp  |
| Roland JD-990                                                    | Internal: 66       | Ghosts           | Dust and Echoes                                         |                                                   | Booe          | Wind ar  |
| Roland JD-990                                                    | B: 68              | Oasis            | Dust and Echoes                                         |                                                   | Booe          | Low drc  |
| E-MU E3 Emulator Standards                                       | Stereo Strings     | Full Strings 88  | Enough Dead Heroes                                      | x20_music01_x20_music03_x20_music04               | Booe          | High "s  |
| Roland XV-5080                                                   | PR-E 065           | St. Choir        | Enough Dead Heroes                                      | x20_music01_x20_music03_x20_music04               | Booe          |          |
| Roland XV-5080                                                   | PR-D 051           | Wind Symphony    | Enough Dead Heroes                                      | x20_music01_x20_music03_x20_music04               | Booe          | *type is |